

HIKET — Data Preparation

Methods and pipeline documentation

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1 Overview

This document describes the data preparation pipeline for the HIKET Bayesian SOC model calibration study. The pipeline is implemented in `Data_work.R` and produces three output files in

Data/model_inputs/:

File	Content	Resolution
input_raw_monthly.csv	Litter inputs + climate	plot $\times$ year $\times$ month
climate_monthly.csv	Climate normals	plot $\times$ year $\times$ month
site_raw.csv	Site attributes + <code>calib_ready</code> flag	plot

The same prepared data feed all six calibrated models (SP1, TP2, TP3, Yasso07, Yasso15, Yasso20; RothC forthcoming). The data preparation is model-agnostic: the simple models (SP1/TP2/TP3) consume the summed litter input, while the Yasso family consumes the AWEN-fractionated inputs, but all draw from the same `input_raw_monthly.csv`, `site_raw.csv`, and the same calibration plot set defined by the `calib_ready` filter (Section 5).

2 Data Sources

2.1 SOC Observations

The SOC data comprise three national soil C campaigns on the Finnish NFI permanent plot network (3,009 plots established in 1985–86 as part of VMI, the Finnish National Forest Inventory). The first two campaigns are read from `Data/SOC/soilC1985_2006.csv`; the third (Komeetta, 2024) is appended from a separate source file in `Data_work.R` section 1.0b (see below):

Campaign 1 — VMI8, 1985–86. The first Finnish national forest inventory permanent plot network, including soil C measurements. Mineral layer protocol: 0–5 cm and 5–20 cm measured separately, plus 20–40 cm. 414 plots with soil data.

Campaign 2 — Biosoil, 2006. European-wide Level I soil monitoring campaign coordinated by ICP Forests / EU. The Finnish contribution measured soil properties including SOC stocks at the same NFI permanent plots (linked via `bio_soil_id` in the NFI plot records). Mineral layer protocol: 0–10 cm and 10–20 cm (different aggregation from 1985), plus 20–40 cm. Same 414 plots as the 1985 campaign.

Campaign 3 — Komeetta, 2024. A new Finnish national soil C campaign on the same permanent plot network, now integrated as the third observation year (`Data_work.R` section 1.0b, source `Data/Komeetta/Komeetta_mitatut_hiilet.xlsx`). Plot identifiers are the `koealatunnus_BIOSOIL` codes directly — no key lookup is needed (627/630 unique VMI codes matched the BIOSOIL key). Layers follow the standard Finnish NFI coding: 101 = H (humus), 102 = F (fermentation), 103 = L (litter) for the organic layer; 201 = 0–5 cm, 202 = 5–20 cm, 203 = 20–40 cm, 204 = 40–60 cm for the mineral layers.

Processing decisions:

- **Organic layer aggregation:** `KOM_ORGANIC_ALL_SUBLAYERS = TRUE` sums the three OFH sublayers (101 + 102 + 103) into a single organic stock. [PLACEHOLDER: confirm with colleagues which sublayers were measured in the 1985/2006 campaigns; the flag switches to H-only (101) if needed to match the earlier protocol.]

- **40–60 cm layer (204):** ingested and stored but not currently used in the 1-m depth extrapolation (which covers 0–40 cm only). Flagged as a possible future constraint on the pooled- λ fit.
- **Quality control:** a weighted-mean SOC check targets the coauthor-reported value of 58 Mg ha⁻¹ (northern plots, $y \geq 7\{.,\}317\{.,\}805$ m ETRS-TM35FIN, weighted $\times 3$ per sampling design). A per-layer plausibility filter drops any single layer exceeding 150 Mg ha⁻¹ as a likely data-entry error; three plots are affected (55652, 3314, 71471), each handled individually (partial-layer retention or exclusion via the organic-missing / single-mineral-layer guards).
- **Anomalous rows:** three rows with non-integer layer codes are dropped.

Komeetta observations are placed at year 2024 (June). Because the MUSTIKKA litter series ends in 2023, `input_raw_monthly.csv` is extended to 2024 by copying each plot’s 2023 litter row (litter is approximately flat inter-annually, so the error is negligible); this keeps the 2024 SOC observation at its correct year label rather than displacing it to 2023.

File structure: after the Komeetta append, `SOC_dedup` spans three observation years (1985, 2006, 2024). The 1985/2006 records come from `soilC1985_2006.csv` (414 plots $\times$ 2 years; each plot in both years). The layer-name difference between the 1985 (0–5/5–20 cm) and 2006 (0–10/10–20 cm) campaigns is handled in `Data_work.R` by accepting both aggregation conventions.

- **Layers accepted:** organic (OFH), 0–5 cm, 5–20 cm, 0–10 cm, 10–20 cm, 20–40 cm.
- **Coverage after matching with litter data:** 412 plots.

Observations are placed in June of the respective measurement year in the monthly input table, consistent with the summer measurement period of the Finnish NFI.

2.2 Litter Inputs

Annual tree litter carbon inputs from Tupek et al. (2026), Zenodo DOI 10.5281/zenodo.19736499.

The dataset integrates NFI permanent plot data from 2805 Finnish sites (inventory rounds 1985, 1990, 1995, 2023) with satellite-derived basal area observations (2005–2023) to produce continuous annual time series 1985–2023. Litter inputs from living biomass turnover and harvest residues are summed to total litter, then distributed across three size classes and four AWEN chemical fractions.

Size classes:

Code	Description
nwl	Non-woody litter (foliage + fine roots)
fwl	Fine woody litter (branches + coarse roots + stem bark)
cwl	Coarse woody litter (stumps)

AWEN fractions: acid-soluble (a), water-soluble (w), ethanol-soluble (e), non-soluble/lignin (n). Fraction proportions are species- and component-specific. Carbon content assumed 50% of dry mass throughout (i.e., $\text{Cha} = 0.5 \times \text{litter dry mass} \times \text{AWEN fraction}$).

- **Files used:**
 - `Data/LitterData/tree_litter_per_site_year_awen_29.04.26.csv` — aggregated across components, resolved by site $\times$ year $\times$ species $\times$ size $\times$ AWEN
 - `Data/LitterData/tree_litter_per_site_year_by_component_29.04.26.csv` — component-level detail (used for diagnostics)

- **Coverage after plot-key matching:** 2794 plots; 412 have both litter and SOC data.
- **Units:** t C ha⁻¹ yr⁻¹.

Plot-key linkage between the three identifier systems (MUSTIKKA, VANHA, BIOSOIL) is performed via `Data/soil_litter_site_key.csv`.

2.3 Climate

Daily gridded meteorological data for all Finnish NFI permanent plots.

- **File:** `Data/Climate/nfi_plot_weather_data_1961_2025.nc`
- **Variables extracted:** mean, min, and max air temperature (°C); precipitation (mm d⁻¹); global radiation (W m⁻²).
- **Period:** 1961–2025.
- **Aggregation:** daily values aggregated to monthly means (temperature) and sums (precipitation, PET).

Potential evapotranspiration (PET) is computed via the Hargreaves–Samani equation:

$$\text{PET} = 0.0023 \times \frac{R_g}{1000} \times \sqrt{T_{\max} - T_{\min}} \times (T_{\text{avg}} + 17.8)$$

where R_g is global radiation (W m⁻² d⁻¹) and temperatures are in °C.

2.4 Soil Properties — Biosoil

2006 European soil monitoring campaign, Finnish subset.

- **Files:** `../.. /Datasets/BioSoil_maaperäaineisto_2006/soil_data_main.csv` and `plot_index.csv`.
- **Mineral topsoil:** layers M01 (0–10 cm) and M12 (10–20 cm) averaged. Variables include texture (clay, silt, sand), bulk density (measured and estimated), pH, exchangeable cations, organic C, total N, organic matter.
- **Organic layer (OFH):** depth and weight.
- **Cajander fertility class (KA):** 1–4 for mineral sites; 11–13 for peatland types (excluded from calibration, see Section 5).
- **Biogeographic zones:** SYKE administrative zones (`kasvyo_syke`, 5 levels south to north) and Ahti (1968) floristic zones (`kasvyo_ahti`).
- **Coverage:** 99.8% of SOC plots matched to Biosoil records.

Derived soil chemistry variables computed in the pipeline:

- `CN_ratio` = OrganicCarbon / TotalNitrogen
- `CEC` = sum of exchangeable cations + free H acidity
- `base_saturation` = (Ca + Mg + K + Na) / CEC

2.5 GTK Soil Classification

Geological Survey of Finland (GTK) 1:1M soil map, queried via ArcGIS REST API (MapServer layer 3, endpoint: `gtkdata.gtk.fi/.../Hasu/maapera/MapServer/3`). Soil codes include mineral classes (Mr, HM, Sa, ...) and non-mineral classes (Tu = peat, Ve = water, Ka = bedrock).

Used for: (1) descriptive soil type field in `site_raw`; (2) per-class pooled estimation in the SOC depth extrapolation. **Not used as a calibration filter** (see Section 5.3).

2.6 NFI Stand Attributes — Kuvio

Stand-level attributes from the Finnish NFI permanent plot records (Pysyvätkoealat), three measurement waves.

- **Files:** `Data/Pysyvätkoealat/{1985,1990,1995}/kuvio.csv`
- **Key variables extracted:** stand age, basal area, mean height, development class, site fertility type (`kasvup_tyyppi`, 8 classes), main site type (`alaryhma`), drainage status (`ojitustilanne`), cutting events (`hakk_laatu`), stand treatments (`metsh_toim`), soil preparation (`aiempi_toim`), temperature sum (`lamposumma`), elevation.
- **Temporal coverage of management indicators:** 1985 wave covers cuttings 10 years back and soil preparation 30 years back; 1990 records new events since 1985; 1995 covers 1990–1995. Combined: approximately 1955–1995.
- **Note on `kasvup_tyyppi`:** this 8-class NFI site fertility system (1 = herb-rich *lehdot* → 8 = fell/subalpine) is more informative than the Cajander class KA (4 mineral classes) for characterising the climate–productivity relationship across the calibration set.

3 SOC Depth Extrapolation to 1 m

Yasso is parametrised for a 1 m soil profile. Measured data cover OFH + mineral 0–40 cm only. The 40–100 cm tail is estimated by an exponential depth model:

$$C(z) = C_0 \cdot e^{-\lambda z}$$

fitted in two stages to balance shape stability with per-profile amplitude flexibility.

Stage 1 — Pooled per GTK soil class. All profiles within a soil class are pooled. C_0 is concentrated out analytically at each candidate via no-intercept OLS (closed-form solution), reducing the optimisation to a 1-D search over λ alone. A global fallback (all classes pooled, 0.026 cm⁻¹) is computed for codes with too few profiles.

Stage 2 — Per-profile C_0 rescaling. Holding the class fixed, C_0 is solved analytically per profile via no-intercept OLS against the observed layer stocks. This anchors the depth shape to the class while allowing each profile its own amplitude.

Layer integral form (fitted directly to observed layer stocks, no midpoint approximation):

$$\text{Stock}(z_1, z_2) = \frac{C_0}{\lambda} (e^{-\lambda z_1} - e^{-\lambda z_2})$$

Total 1 m SOC = OFH stock (measured, added directly) + mineral 0–40 cm (measured, summed) + mineral 40–100 cm (extrapolated). The raw 0–40 cm + OFH sum is retained as `soc_obs_tCha_sum` for diagnostic purposes. Of 824 plot-year combinations attempted, 822 were successfully extrapolated (2 lacked sufficient mineral layer data and fall back to the raw sum).

4 Organic Layer Quality Filters

Two organic layer flags are computed from the SOC data and propagate into the `calib_ready` filter.

- **organic_missing:** OFH layer never recorded in any measurement year. Indicates a structural data gap; the 1 m extrapolation cannot be performed without an OFH stock. Affects 12 plots.
- **organic_zero:** OFH recorded but $C_{kgha} = 0$ in at least one year. Likely a measurement issue rather than genuine absence; excluded out of caution. Verified non-overlapping with `organic_missing`. Affects 28 plots.

5 Calibration-Ready Plot Filtering

The `calib_ready` flag in `site_raw` defines the calibration set. A plot is included if and only if all of the following hold:

#	Criterion	Rationale
1	<code>has_climate</code>	Plot present in climate netCDF with non-NA temperature
2	<code>has_soc</code>	At least one SOC observation in <code>SOC_agg</code>
3	<code>!peatland</code>	Cajander KA {11, 12, 13}; peatland C dynamics not represented by Yasso
4	<code>!organic_missing</code>	OFH never recorded; extrapolation impossible
5	<code>!organic_zero</code>	OFH recorded as zero in any year; excluded as data quality issue
6	<code>!high_mrt</code>	Implied MRT > 100 years (see below)

Implied MRT filter (criterion 6). Implied MRT = $\text{mean}(1 \text{ m SOC}) / \text{mean}(\text{annual litter input})$. Plots with MRT > 100 years have litter inputs so low that the model predicts near-zero SOC at any parameter value, causing a $-\infty$ likelihood at the pre-MCMC sanity check. The 100-year threshold corresponds to a clear gap in the MRT distribution. This filter also catches four plots with zero litter after AWEN-compound mapping (MRT = ∞) that bypass criterion 5, which operates on raw litter inputs before compound-fraction assignment. In total, 14 plots are excluded (4 with MRT = ∞ , 10 with MRT 100–489 years); several overlap with GTK peat/water classes, consistent with peat-influenced carbon dynamics unsupported by Yasso’s mineral-soil parameterisation.

GTK soil class — reviewed, not used as a filter. Tu (peat), Ve (water), and Ka (bedrock) classes together constitute ~26% of plots passing criteria 1–6. Inspection showed: (a) SOC distributions for these classes are statistically indistinguishable from mineral plots (Tu median 59, Ve 59, Ka 70, mineral 62 Mg C ha⁻¹); (b) Cajander vegetation surveys classified 53 of 96 such plots as KA = 1 (rich mineral upland forest), with none classified as peatland (KA 11–13). The disagreement is

consistent with the GTK map’s coarse resolution (1:1,000,000). GTK class is retained in `site_raw` for descriptive use and for per-class estimation in the depth extrapolation.

Final calibration set: 351 plots (365 before the MRT filter; 350–351 after matching with climate and SOC records).

6 Derived Site Covariates (`site_raw`)

The following covariates are computed and stored in `site_raw` for use in the residual random-forest analysis.

6.1 Climate Normals and Variability

Computed from long-run monthly normals (all available years in the netCDF):

Variable	Description
<code>mean_temp</code>	Annual mean air temperature (°C)
<code>mean_precip_annual</code>	Annual precipitation sum (mm yr ⁻¹)
<code>PET_annual</code>	Annual Hargreaves–Samani PET sum
<code>coldest_month_T</code> , <code>warmest_month_T</code>	Monthly temperature extremes
<code>T_seasonality</code>	SD of monthly temperature normals
<code>P_seasonality</code>	CV of monthly precipitation normals
<code>GDD5</code>	Growing degree days base 5 °C (weighted by calendar-month length)
<code>aridity_index</code>	<code>mean_precip_annual</code> / <code>PET_annual</code>
<code>koppen_class</code>	Köppen–Geiger class (Beck et al. 2018, simplified; Dfb or Dfc for Finland)
<code>temp_zone</code>	Tercile binning of <code>mean_temp</code> into cold / intermediate / warm

6.2 Litter Quality

Computed from cumulative litter totals across the full time series:

Variable	Description
<code>litter_{A,W,E,N}_frac</code>	AWEN compound fractions of total plot litter
<code>woody_share</code>	(<code>fwl</code> + <code>cwl</code>) / (<code>nwl</code> + <code>fwl</code> + <code>cwl</code>)
<code>conifer_share</code>	<code>sp_frac_1</code> (Scots pine) + <code>sp_frac_2</code> (Norway spruce)

All three size classes (`nwl`, `fwl`, `cwl`) are sourced directly from the Tupek et al. dataset.

6.3 Soil Chemistry

Derived from Biosoil mineral topsoil (M01 + M12 layers, averaged):

Variable	Description
CN_ratio	OrganicCarbon / TotalNitrogen (NA where N = 0)
CEC	Sum of exchangeable cations + free H acidity
base_saturation	(Ca + Mg + K + Na) / CEC

For RF covariate selection: prefer **EstimatedBulkDensity** over **MeanBulkDensity** (the latter has 86% NA in the calibration set). Avoid using **OrganicCarbon** and **OrganicMatter** from Biosoil as RF covariates — they are highly collinear with the SOC observations and would trivially recover the SOC signal rather than model residual structure.

7 Output Files

7.1 input_raw_monthly.csv

Monthly time series, one row per (plot_id, year, month). Coverage spans 1985–2024 (litter 1985–2023 from MUSTIKKA, plus a 2024 row copied from 2023 to host the Komeetta observation), 12 months/year. [PLACEHOLDER: exact plot count and total row count after the Komeetta extension — re-run `Data_work.R` to confirm; pre-Komeetta this was $405 \text{ plots} \times 39 \text{ years} \times 12 = 189,540$ rows.]

Litter inputs in $\text{t C ha}^{-1} \text{ month}^{-1}$ (annual values divided by 12, i.e. assuming uniform within-year distribution). SOC observations placed in June of the observation year. Plots without full climate coverage are excluded; SOC observations for excluded plots are not carried forward.

7.2 climate_monthly.csv

Standalone monthly climate per plot, deduplicated. Same plot coverage as `input_raw_monthly.csv`.

7.3 site_raw.csv

Comprehensive per-plot attributes. Columns cover all covariates listed in Section 6 plus all raw Biosoil, GTK, and kuvio variables, exclusion flags (**peatland**, **organic_missing**, **organic_zero**, **high_mrt**), and the **calib_ready** flag. [PLACEHOLDER: total plot count and **calib_ready** count after Komeetta integration — pre-Komeetta these were 419 total / 351 calib-ready; the Komeetta append adds a third observation year and may shift the organic-layer flags and final counts. Re-run `Data_work.R` to confirm.]

8 Automated Sanity Checks

The script ends with 25 automated checks. Key checks relevant to the calibration:

- No duplicate plot–year–month rows
- No NA in climate fields
- All plots have the same month count (the check is generic; with the 1985–2024 span this is $40 \text{ years} \times 12 = 480$ rows per plot)
- No negative litter values

- SOC observations placed only in June
 - SOC count consistent between `input_raw_monthly` and `SOC_agg`
 - Litter magnitude median within boreal range (1.5–4.5 t C ha⁻¹ yr⁻¹; observed median 2.56 after CWL correction)
 - SOC plausibility for calib-ready plots (5–400 t C ha⁻¹)
 - 1 m extrapolated median raw sum median
 - No peatland, organic-excluded, or high-MRT plots in `calib_ready`
 - All derived covariates present in `site_raw`
 - AWEN fractions sum to 1 per plot
-

9 Software and Reproducibility

Language: R (4.3).

Key packages: `sf`, `ncdf4`, `httr`, `jsonlite`, `rnaturalearth`, `dplyr`, `ggplot2`.

HPC: Production runs on CSC Puhti (Finland); development on macOS.

To reproduce from scratch, run `Data_work.R` from the project root directory. The GTK soil type extraction (section 1.3) requires internet access to the GTK ArcGIS REST API; all other steps are self-contained given the data files. After running, sync outputs to Puhti and recompile the Fortran shared libraries before submitting calibration jobs (see `Calibration_real_data/run_hiket_pipeline.R` for the full workflow).